Block-Diffusion OD Planning — Consolidated Results (v3)

Status (2026-07-20, v3). Every number in this document is re-measured on the **v2 backbones** (14 models retrained with seeded 80/20 splits: mask×7, graph d=2.0×7). Evaluation: the normal task uses a **shared held-out 1,000-pair set** drawn from the untouched test split (§2: 3 generation seeds, mean±std); the exceptional task uses the except_0 **reserved 1,000 pairs** excluded from discriminator training. The developmental narrative and full ablation history live in `RESULTS_BD_v1.pdf` (archived); full derivations in `BD_GUIDANCE_FORMULATION.pdf` , `BD_CODE_VERIFICATION_KO/EN.pdf` , `BD_ARCHITECTURE_KO.pdf` . v1 checkpoints/discs archived as `v1_*` .

1. Setup and architecture

BD3-LM-style block diffusion (semi-AR across blocks, discrete diffusion within), 6-layer / dim-256 DiT (adaLN), porto_v3_normal (1,390 vertices, 1.93M paths), 1 epoch = 48.2k iters (bs 32, lr 5e-4). Canvas `[dst, ori, v…, v_L=dst, END…| PAD…]` : the END tail inside the dst block is supervised, PAD blocks excluded. Kernels: mask (absorbing, SUBS, first-hitting — max data length 59, so blk64 is effectively MDLM) and graph (uniform-state CTMC, d = 2.0, T = 100). Generation is open-ended (dst→hit / END→miss).

1.1 Detailed configuration

Denoiser	DiT 6 layers / dim 256 / heads 8 / cond 128 / dropout 0.1, adaLN (zero-init), learned absolute positions (shared by halves), node2vec (100d) embedding init
Training	1 epoch = 48.2k iters, bs 32, Adam lr 5e-4, grad clip 1.0, condition dropout 0.1 (O·D nulled jointly); mask: per-block t~U[1e-3,1]; graph: T=100, β∈[1e-4,10], d=2.0
Data / splits	porto v3-0.05 normal: 1,390 vertices, 1.93M shortest paths (max length 59). Deterministic order (shuffle=False) + torch-seed-1 80/20 random_split → fully reproducible membership. Eval: shared 1,000 pairs drawn (seed 777) from the test split
Exceptional data	v4-0.05 except_0…99 (5% of edges removed; normal graph identical to v3). Disc training = 1% (19.3k paths) after excluding the reserved rows; evaluation = the reserved 1,000 pairs (never seen by the disc)
Discriminator (IW)	clean random-truncation partial-path input; 2-layer/dim-128 transformer + 2-layer GraphSAGE + transition features [edge-existence, is-transition, deg-ratio] + masked mean pooling + bounded logit λ=4; BCE (label smoothing 0.05), Adam 1e-3, 4,000 steps, bs 128. Negatives = 20k unconditional generations of the very block's v2 model
D-CBG classifier	identical architecture (+MASK token, t-conditioned embedding); input = generation-time-corrupted noisy states (their protocol); same 1% data, same negative pools (noised their way)
Guidance	N = 100 mean-field candidates per reveal, w = exp(bounded logit) = D/(1−D), applied at every reveal, self-normalized, ESS logged; adj-proposal: A_scn masking w.r.t. revealed neighbours (Lemma 3); D-CBG: γ = 4 (γ ≤ 2 inert)
Hardware	RTX 3090 (24GB) ×4, torch 1.12.1+cu113

1.2 Performance metrics

hit	fraction of RAW model outputs that reach dst unaided; always measured pre-refinement, so post-processing can never inflate it (hit = arrival in raw rows)
arrival	fraction of final paths (after the row's post-processing) whose last vertex equals dst
valid	fraction of paths whose EVERY consecutive pair is an edge of the evaluation graph (all-or-nothing per path; against $A_{except0}$ in exceptional runs) — small edge error rates get amplified by path length (§2.3)
valid∧hit (raw)	valid AND hit — the post-processing-proof usable-path rate; equals raw hit under P1+P3 (the one remaining model-dependent quantity)
EM / PC	repo-standard scoring (eval_shortest.evaluate_em_pc): EM = exact match rate against the OD pair's reference shortest path; PC = partial coverage (overlap) of the reference path's vertices; both averaged over paths
DTW	dynamic time warping distance between generated/reference (lat,lng) sequences (L1, ~km scale), normalized by max length — geometric trajectory similarity
len err / len corr	mean |generated − real length| / Pearson correlation of the two — calibration of the emergent length (§2.3: dominated by dst-truncation geometry on hits)
inv-edge / rem-edge / inv-node	invalid-edge rate (edge level) / the removed-edge portion (edges present in normal but closed in the scenario) / fraction of vertices touching an invalid edge
ESS	effective sample size of the importance weights per reveal, $(\Sigma w)^2/\Sigma w^2 \in [1, N]$. ≈N means flat weights (inert guidance); ≈1 means weight collapse — the dial separating the two failure modes
patch	average number of tokens inserted by post-processing (P1 bridges + P3 grafts)
wall-clock (s/path)	measured **wall-clock time to generate one path**: elapsed time of the generation loop (model forwards + guidance computation; data loading/scoring excluded) ÷ number of paths, on one idle RTX 3090, batch 100, 200–1,000 pairs. Unlike classifier CALL COUNTS it reflects

	batching efficiency, backward-pass costs and sequence lengths — reported only for the D-CBG vs IW comparison (§5)

Figure 1. Block-diffusion mechanics — and the pathway of prior-block information (code details & line numbers: `pdfs/BD_ARCHITECTURE_KO.pdf`)

A. Canvas and length spec (blk4 example; `build_canvas`)

`dst` `ori` `v₁` `v₂` `v₃` `v₄` `v₅` `v₆=dst` `END` `END` `PAD` `PAD`

Yellow = clean conditioning prefix (out of loss); the END tail inside the dst block IS a training target; PAD blocks are excluded.

B. Prior-block information enters through self-attention alone

	B1	B2	B3	B1′	B2′	B3′
B1	BD	·	·	·	·	·
B2	·	BD	·	OBC	·	·
B3	·	·	BD	OBC	OBC	·
B1′	·	·	·	BD	·	·
B2′	·	·	·	BC	BD	·
B3′	·	·	·	BC	BC	BD

	B1	B2	B3*
B1	✓	·	·
B2	✓	✓	·
B3*	✓	✓	✓

Left (training, `get_block_diff_mask` **)**: on the $[x_t(B1–B3) \| x_0(B1′–B3′)]$ input, noised block B3 attends to itself (BD) and to the **clean x_0 of strictly earlier blocks** (**OBC** — the sole training-time channel for prior-block information). BC maintains the clean half's representations.

Right (sampling, `get_block_causal_mask` **)**: every token of the current block B3* attends **directly, at every layer**, to committed B1, B2 (orange). Committed tokens are frozen (t=0), re-encoded each forward (no KV cache).

The adaLN conditioning path (c = t + OD embedding) carries no prior-block content — content flows 100% through attention.

C. The BDTransformer tower — read bottom-up; exact injection points of conditioning and prior-block content

$\hat{x}_0$ logits (b, S, V+3) — per-position x_0 prediction
↑
zero-init Linear head
↑
final scale & shift (γ, β only, no gate)
↑
final LayerNorm
↑

BDBlock ×6 (bottom→up)
⊕ residual 2
↑
gate $\times\alpha_2$
↑
Pointwise MLP (4×, GELU)
↑
scale & shift: $\times(1+\gamma_2) + \beta_2$
↑
LayerNorm (affine-free)
↑
⊕ residual 1
↑
gate $\times\alpha_1$
↑
Multi-Head Attention (block-causal mask)
Q ← current-block tokens | **K/V ← all visible tokens incl. committed blocks**
= the only entry point of prior-block content
↑
scale & shift: $\times(1+\gamma_1) + \beta_1$
↑
LayerNorm (affine-free)

↑
+ pos_emb[arange(S) % n] (positions shared by the two halves)
↑
in_proj (100→256) ← x_embedding (node2vec init)
↑
tokens: x_- (sampling: [committed ‖ current block]) / training: $[x_- \| x_0]$

conditioning path — coefficients only, no content

t_tok (per-block noise level; committed t=0) → time_mlp
(ori, dst) → emb → od_mlp (null_cond on drop)
↓ +
c (b, S, 128)
↓
adaLN MLP (Mish→Linear, **zero-init**)
→ $(\gamma_1,\beta_1,\alpha_1,\gamma_2,\beta_2,\alpha_2)$

adaLN feeds every γ/β/α box in every block
↖ all orange boxes

- **γ, β, α are NOT model parameters.** The learned parameters are the purple adaLN MLP's weights (zero-init → every block starts as identity); γ/β/α are **activations computed from c per token and per sample**. Conditioning never touches weights — it **scales (1+γ)/shifts (β) the normalized activations feature-wise and gates (α) sublayer outputs** (why the LayerNorms are affine-free).
- **Prior-block content enters through the attention K/V** (blue box), not through adaLN. In short: **content via K/V; conditioning (per-block t, OD) via γβα.**

```
(γ₁,β₁,α₁,γ₂,β₂,α₂) = adaLN_MLP(c).chunk(6)          # BDBlock.forward, bd_models.py:149–153
x = x + α₁ ⊙ Attn( (1+γ₁)⊙LN(x) + β₁ , attn_mask )
x = x + α₂ ⊙ MLP ( (1+γ₂)⊙LN(x) + β₂ )
```

D. Open-ended semi-AR generation: append block → first-hitting reveals one token per forward (each revealed token immediately becomes context) → scan the committed block: dst → hit (stop) / END → miss (stop) / else append the next noised block.

Training's OBC (noised → clean past) corresponds exactly to sampling's committed-block reference (teacher forcing), so the conditioning distributions match. At blk1 all information flows through committed references (a causal LM); at blk64 (= MDLM) OBC is empty — block size interpolates the ratio of the two channels (§6.9).

2. Block-size sweep — held-out, 3 seeds mean±std

Post-processing status: none. Every metric in §2 (§2.1, §2.2, §2.3) is RAW model output — no P1/P3, no refinement of any kind. Post-processing appears only from §3 onward.

2.1 mask

blk	hit (=arr)	valid	valid∧hit (raw)	EM	PC	DTW	len err	len corr
1	0.997±.002	0.934±.007	0.934±.007	0.830±.011	0.869±.010	0.032	0.84	+0.864

2	0.958±.007	0.842±.006	0.822±.006	0.769±.005	0.799±.005	0.038	0.45	+0.982
4	0.958±.001	0.836±.004	0.811±.004	0.747±.010	0.783±.008	0.037	0.37	+0.991
8	0.951±.002	0.808±.006	0.774±.006	0.716±.008	0.754±.007	0.042	0.40	+0.993
16	0.898±.006	0.680±.006	0.625±.002	0.607±.009	0.637±.008	0.061	0.56	+0.989
32	0.858±.010	0.474±.004	0.420±.002	0.413±.002	0.437±.002	0.081	0.96	+0.968
64	0.839±.005	0.430±.014	0.373±.010	0.366±.012	0.391±.012	0.072	0.89	+0.988

2.2 graph (d = 2.0)

blk	hit (=arr)	valid	valid∧hit (raw)	EM	PC	DTW	len err	len corr
1	0.429±.001	1.000±.000	0.429±.001	0.093±.006	0.168±.007	0.437	36.4	+0.237
2	0.173±.014	0.997±.001	0.173±.014	0.399±.008	0.558±.005	0.488	6.48	+0.492
4	0.128±.007	0.978±.002	0.128±.007	0.309±.012	0.499±.008	0.302	4.50	+0.719
8	0.058±.004	0.991±.001	0.058±.004	0.124±.012	0.331±.006	0.282	4.53	+0.783
16	0.029±.002	0.996±.002	0.029±.002	0.148±.008	0.390±.006	0.377	5.66	+0.705
32	0.035±.002	0.969±.002	0.035±.002	0.174±.002	0.411±.004	0.394	5.96	+0.693
64	0.027±.004	0.966±.003	0.027±.004	0.198±.006	0.430±.001	0.363	5.74	+0.724

2.3 Failure anatomy (v3, 1,000 held-out pairs, one fixed generation seed = 7, RAW)

Post-processing status: none. Same settings as §2.1/2.2 — both kernels, all 7 block sizes — on the identical held-out 1,000 pairs, a single fixed seed (no seed-averaging), raw outputs. "boundary/interior" = which invalid edges land on a block-first token (predicted from the prior block, AR-like) vs a within-block token.

mask	hit	len corr (hit)	len corr (miss, n)	miss→dst med	miss→dst mean	≤2 hops	inv-edge	boundary/interior
1	0.997	0.972	– (3)	14	13.0	0%	1.02%	234 / 0
2	0.956	0.989	0.962 (44)	2	2.41	86%	1.70%	192 / 189
4	0.954	0.987	0.976 (46)	2	1.93	87%	1.03%	35 / 196
8	0.954	0.995	0.974 (46)	2	3.22	89%	1.06%	25 / 211
16	0.909	0.992	0.959 (91)	2	3.32	89%	1.98%	12 / 429
32	0.871	0.963	0.905 (129)	2	2.95	84%	4.27%	7 / 920
64	0.853	0.991	0.974 (147)	2	3.67	84%	4.54%	0 / 1001

graph d=2.0	hit	len corr (hit)	len corr (miss, n)	miss→dst med	miss→dst mean	≤2 hops	inv-edge
1	0.432	0.458	0.054 (568)	9	13.5	5%	0.00%
2	0.194	0.902	0.463 (806)	4	10.7	31%	0.00%
4	0.119	0.969	0.723 (881)	4	6.93	34%	0.00%
8	0.072	0.978	0.759 (928)	5	7.82	26%	0.00%
16	0.032	0.996	0.702 (968)	5	8.40	19%	0.00%
32	0.029	0.997	0.703 (971)	6	9.39	17%	0.00%
64	0.027	0.994	0.729 (973)	6	8.96	17%	0.00%

- **Mask misses are near-misses**: for blk2–64 the miss endpoint sits a median 2 hops from dst (84–89% ≤2 hops) — a last-token identification slip that P3 endpoint-patch repairs cheaply. The sole exception is blk1: only 3 misses, but they run far (median 14 hops, 0% ≤2) — rare pure-AR runaways. Length calibration holds for hits (0.96–1.00) and misses (0.90–0.98) throughout.
- **Validity decay = within-block interior skips, now monotone across the whole sweep**: as block size grows the invalid edges migrate from block-boundary to block-interior — blk2 192/189 → blk4 35/196 → blk16 12/429 → blk64 0/1001 (strictly interior). blk1 is all-boundary by definition (every token is a block start). Block-boundary (AR-like) transitions stay clean; the cost is paid strictly **inside** blocks, from first-hitting's random within-block reveal order. Edge-error rate climbs only 1.0%→4.5% (median 2–3-hop skips), but per-path all-or-nothing validity amplifies it.
- **The graph kernel is the mirror image — and its misses are NOT near-misses**: constrained decoding holds inv-edge at 0.00% for every block size, but hit collapses 0.43→0.03 as the block grows. Unlike mask, graph miss endpoints land far from dst (median 4–9 hops, mean 7–13.5, only 5–34% ≤2) — and the mean sitting well above the median exposes a heavy right tail (many misses drift very far), whereas mask misses have mean≈median≈2 (tight, no tail). Genuine navigation drift, not a last-token slip, so endpoint patching cannot repair them cheaply. Block size charges mask in validity, graph in arrival.

- **Held-out validation**: v2 numbers match v1 (train-overlapping) within noise — no memorization inflation. With 1.93M paths and 6.8M parameters (~3.5 params/path) the model cannot memorize verbatim; what it learns — local transition choices toward an anchored destination — transfers unchanged to unseen OD pairs.
- **Untested note**: the END tail's loss share grows as (B−1)/2 — ~58% of supervised tokens at blk64 — leaving a loss-engineering lever (constant-mass reweighting / first-END-only; v1 §4.5).

2.4 Reveal order — first-hitting vs left-to-right (mask, RAW unless P1P3, 1,000 held-out, seed 7)

Controlled ablation of the within-block reveal order. The mask sampler reveals one masked position per step; the default picks it **uniformly at random** (first-hitting), here compared against a deterministic **left-to-right** (l2r) order. Everything else — the first-hitting *time* schedule, logits, seed — is identical, so this isolates reveal order only. Notation `fh → l2r`; **bold** marks an l2r gain ≥2%p.

blk	raw valid (fh→l2r)	raw EM (fh→l2r)	raw arrival (fh→l2r)	P1+P3 EM (fh→l2r)
1	0.934 → 0.930	0.832 → 0.825	0.997 → 0.995	0.849 → 0.841
2	0.852 → 0.851	0.779 → 0.771	0.956 → 0.958	0.804 → 0.798
4	0.823 → **0.899**	0.734 → **0.785**	0.954 → 0.970	0.768 → **0.801**
8	0.827 → **0.883**	0.728 → **0.785**	0.954 → 0.966	0.768 → **0.798**
16	0.705 → **0.817**	0.617 → **0.710**	0.909 → **0.947**	0.662 → **0.720**
32	0.462 → **0.695**	0.397 → **0.572**	0.871 → **0.896**	0.503 → **0.601**
64	0.441 → **0.735**	0.379 → **0.600**	0.853 → **0.891**	0.533 → **0.617**

- **Left-to-right wins, and the gain grows with block size**: raw validity blk4 +7.6%p, blk16 +11.2%p, blk32 +23.3%p, blk64 **+29.4%p** (0.441 → 0.735); raw EM blk64 +22.1%p (0.379 → 0.600). Arrival also rises (blk64 0.853 → 0.891).
- **blk1/2 are unchanged** (≤0.8%p) — a block of 1–2 positions has no reveal-order freedom, the expected sanity check.
- **Direct confirmation of the §2.3 anatomy**: mask validity decay was diagnosed as within-block *interior* skips caused by first-hitting's random reveal order (a token committed before its left neighbour exists). L2R commits each token conditioned on its already-revealed left neighbour (block-interior AR), so consecutive-edge validity is respected — exactly the failure mode L2R removes, hence the largest blocks (most interior positions) benefit most.
- **The gain survives post-processing**: P1+P3 EM blk64 0.533 → 0.617, blk32 0.503 → 0.601, and l2r needs less patching (blk64 avg patch 3.48 → 2.90). L2R is a free, hyperparameter-free change that makes large-block mask sampling competitive again.

3. Post-processing: P1 (local splice) · P3 (endpoint patch)

P1 inserts a shortest-path bridge at every invalid pair (u,v) → valid = 1 by construction, hit untouched. P3 (= refine_nx) grafts a shortest path from a miss endpoint to dst → arrival = 1. hit is always measured pre-refinement (valid∧hit (raw)) — the one model-dependent quantity left under P1+P3.

blk	raw arr	raw valid	valid∧hit (raw)	raw EM	raw PC	P1+P3 EM	P1+P3 PC	patch
1	0.997	0.934	0.934	0.832	0.871	0.849	0.898	1.0
2	0.956	0.852	0.826	0.779	0.810	0.804	0.869	0.5
4	0.954	0.823	0.798	0.734	0.771	0.768	0.846	0.8
8	0.954	0.827	0.794	0.728	0.769	0.768	0.848	0.7
16	0.909	0.705	0.647	0.617	0.651	0.662	0.766	1.9
32	0.871	0.462	0.408	0.397	0.422	0.503	0.642	3.3
64	0.853	0.441	0.390	0.379	0.403	0.533	0.661	3.5

- P1+P3 gives the structural **valid = arrival = 1** guarantee at every block size, with usable-path rate = raw hit. At large blocks P1 also lifts EM/PC (blk64 EM 0.379→0.533) — the bridges are real route segments. Zero unfixable pairs.

4. Importance-weight guidance (exceptional scenario except_0)

Math summary (full: BD_GUIDANCE_FORMULATION.pdf). Guided reverse of block b: $q(x^b_{t-1}|x^b_t, x^{<b}) = \mathbb{E}_{x_0\sim\hat{p}}[w\cdot q(\cdot|x_0)]/\mathbb{E}[w]$, $w = q(x_0|x^{<b})/p_\theta(x_0|x^{<b}) \approx D/(1-D)$. **Lemma 1**: $w = C^{-1}\cdot r(x^{<b}\|x_0)$ with $C = q(x^{<b})/p_\theta(x^{<b})$ candidate-independent, cancelling under self-normalization — a **clean partial-path** discriminator suffices. **Lemma 2**: a block-only classifier learns the marginal ratio and is biased (non-constant mismatch). **Lemma 3**: masking the proposal to A_{scn}-legal transitions w.r.t. revealed neighbours is exact — excluded candidates lie outside supp(q) (w = 0) and Z_ℓ is candidate-common, cancelling (unbiased variance reduction). Implementation: N = 100 mean-field candidates per first-hitting reveal, w = exp(bounded logit), applied at every reveal, ESS logged. Discriminator: GraphSAGE + edge-existence channels (scenario adjacency as input → zero-retraining scenario transfer); negatives = **20k unconditional generations of the very block's model** (the true denominator p_θ; the data<mix<model ordering: v1 §6.7).

4.1 Seen (disc trained on except_0, 1%)

Post-processing: columns without a "P1P3" tag (base/IW/adj+IW *valid* and *EM*, rem-edge, ESS) are RAW; only the explicit `P1P3 EM/PC` columns have P1+P3 applied.
Data (seen): 19,308 positives (except_0, 1%) vs 20,000 model-generated negatives (≈1:1 balanced); base rows (blk1/2) are guidance-free.

blk	base valid	base EM	IW valid	IW EM	IW P1P3 EM	IW P1P3 PC	ESS	adj+IW valid	adj+IW EM	rem-edge	adj+IW P1P3 EM	adj+IW P1P3 PC
1	0.306	0.283	—	—	—	—	—	—	—	—	—	—
2	0.264	0.244	—	—	—	—	—	—	—	—	—	—
4	0.266	0.240	0.331	0.298	0.309	0.464	99.1	0.390	0.338	3.0%	0.357	0.496
8	0.231	0.212	0.298	0.264	0.294	0.453	98.8	0.346	0.298	3.2%	0.335	0.483
16	0.222	0.198	0.277	0.240	0.268	0.423	98.2	0.327	0.270	3.1%	0.295	0.447
32	0.160	0.132	0.225	0.184	0.216	0.376	97.5	0.255	0.203	3.2%	0.236	0.398
64	0.141	0.129	0.201	0.160	0.211	0.372	95.4	0.229	0.176	3.1%	0.230	0.394

4.1b Negative source: model-generated vs normal-data (identical disc architecture, e0, raw)

The §4.1 numbers above use **model-negative** discriminators (negatives = the block's own unconditional generations → the density ratio q_{except}/p_θ, the theoretically correct denominator). Here the same protocol is re-run with a **data-negative** discriminator (negatives = **normal** real paths → it learns q_{except}/q_{normal}, a scenario-shift ratio, not q/p_θ). Both cells are RAW except the explicit P1P3 column; same v2 backbones, same except_0 reserved 1,000 pairs, seed 7.

blk	base valid	IW valid (model / data)	adj+IW valid (model / data)	adj+IW arrival (model / data)	adj+IW EM (model / data)	adj+IW PC (model / data)	adj+IW rem-edge (model / data)	adj+IW P1P3 EM (model / data)
1	0.306	—	—	—	—	—	—	—
2	0.264	—	—	—	—	—	—	—
4	0.266	0.331 / 0.320	0.390 / 0.375	0.898 / 0.904	0.338 / 0.333	0.360 / 0.350	3.0% / 2.9%	0.357 / 0.349
8	0.231	0.298 / 0.285	0.346 / 0.332	0.909 / 0.914	0.298 / 0.288	0.317 / 0.306	3.2% / 3.2%	0.335 / 0.334
16	0.222	0.277 / 0.258	0.327 / 0.308	0.849 / 0.856	0.270 / 0.257	0.293 / 0.277	3.1% / 3.1%	0.295 / 0.294
32	0.160	0.225 / 0.194	0.255 / 0.235	0.760 / 0.767	0.203 / 0.183	0.224 / 0.203	3.2% / 3.2%	0.236 / 0.227
64	0.141	0.201 / 0.182	0.229 / 0.211	0.795 / 0.795	0.176 / 0.168	0.198 / 0.185	3.1% / 3.1%	0.230 / 0.225

- **Arrival and PC**: adj+IW arrival is essentially identical for the two negatives (~0.76–0.91, model≈data), while PC is uniformly (slightly) higher for model (e.g. blk64 0.198 vs 0.185, blk16 0.293 vs 0.277) — the model-negative edge shows up in route quality (PC/EM), not in reaching the destination.
- **Both negatives give real gains; model-negative is uniformly (but modestly) better**: over base, data-negative IW lifts valid +5.4%p at blk4 (0.320 vs 0.266), model-negative +6.5%p (0.331). The gap holds at every block size and for adj+IW (blk64: model 0.229 vs data 0.211). This empirically confirms the data
- **Why model-negative wins theoretically**: guidance needs $w = q_{except}(x_0|x^{<b})/p_\theta(x_0|x^{<b})$. Model-negatives make the discriminator estimate exactly this (denominator = p_θ). Data-negatives estimate q_{except}/q_{normal} — the right *direction* (avoid normal-only routes) but the wrong denominator, so the tilt is slightly miscalibrated.
- **rem-edge is negative-agnostic**: adj+IW removed-edge usage is ~3% for both (2.9–3.2%) — because that reduction comes from Lemma 3 adjacency masking, not from the discriminator's negative source.

4.2 Unseen (disc trained on except 1–99 → evaluated on except_0)

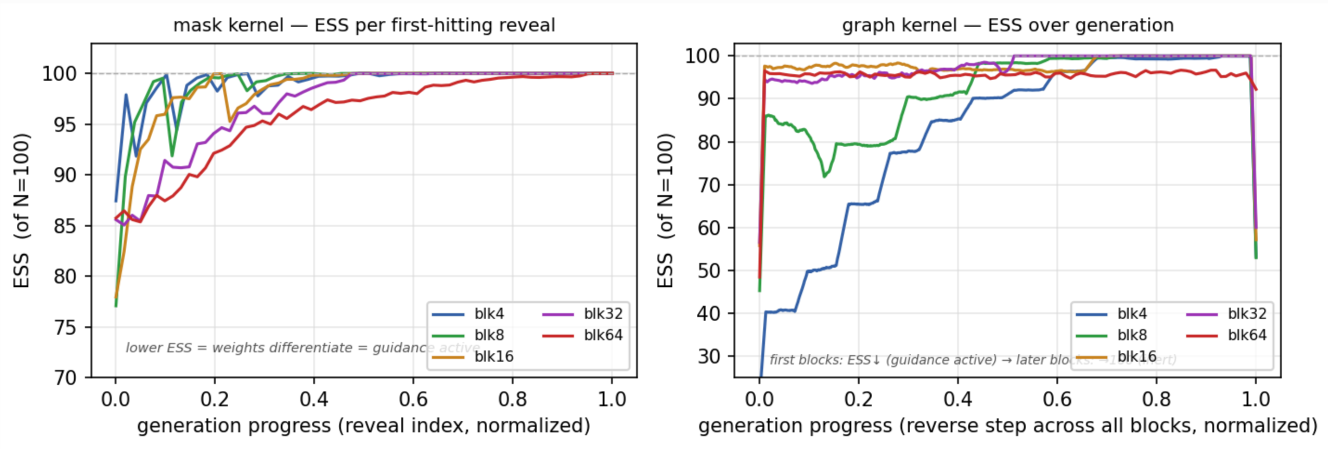
Negatives = model-generated (not normal data). The unseen discriminator is trained on except 1–99 (positives) vs each block's own unconditional generations (negatives = p_θ), adjacency-conditioned per scenario, then evaluated zero-shot on except_0. blk1/2 are **base-only** (guidance not applied). The *seen* column repeats the §4.1 model-negative numbers. **Data counts** — §4.1 (seen): 19,308 positives (except_0, 1%) vs 20,000 model negatives (≈1:1). §4.2 (unseen): 1,911,492 positives (1% × 99 scenarios) vs 20,000 shared model negatives. Crucially, each training step compares **one scenario's** ~19,308 positives against the 20,000 negatives with that scenario's adjacency as input, so every conditional comparison is ≈1:1 — the 96:1 union ratio is never seen by any gradient step, and negatives sample a **single fixed distribution** p_θ that 20k characterizes well. **Imbalance control**: retraining blk16 with a 10× larger negative pool (200,000) moves IW valid 0.277→0.279, adj+IW valid 0.326→0.329, adj+IW P1P3 EM 0.299→0.301 (all ≤0.3%p, within noise) — confirming the count imbalance is harmless.

blk	base valid	IW valid (unseen/seen)	adj+IW valid (unseen/seen)	adj+IW arrival (unseen/seen)	adj+IW EM (unseen/seen)	adj+IW PC (unseen/seen)	adj+IW rem-edge (unseen/seen)	adj+IW P1P3 EM (unseen/seen)
1	0.306	—	—	—				
2	0.264	—	—	—				
4	0.266	0.329 / 0.331	0.390 / 0.390	0.891 / 0.898	0.339 / 0.338	0.360 / 0.360	2.95% / 2.96%	0.356 / 0.357
8	0.231	0.298 / 0.298	0.342 / 0.346	0.913 / 0.909	0.300 / 0.298	0.317 / 0.317	3.27% / 3.21%	0.340 / 0.335
16	0.222	0.277 / 0.277	0.326 / 0.327	0.852 / 0.849	0.273 / 0.270	0.294 / 0.293	3.08% / 3.09%	0.299 / 0.295
32	0.160	0.218 / 0.225	0.249 / 0.255	0.761 / 0.760	0.198 / 0.203	0.219 / 0.224	3.19% / 3.19%	0.237 / 0.236
64	0.141	0.199 / 0.201	0.224 / 0.229	0.798 / 0.795	0.178 / 0.176	0.197 / 0.198	3.06% / 3.05%	0.233 / 0.230

- **Division of labour**: removed-edge avoidance comes from Lemma 3 masking (rem-edge ~5%→3%), route quality from discriminator reweighting (+5–7%p valid). ESS ≈ 95–99: per-step nudges are small but accumulate over reveals. The cost is arrival (–5 to –7%p) — restored structurally by P3.
- **Generalization gap ≈ 0** (blk4 adj+IW 0.390/0.390; gap ≤0.6%p across blk4–64): scenario information enters through the adjacency input, not memorization. The full stack (adj+IW+P1P3) reaches EM 0.357 / PC 0.496 at blk4 — +32%/+15% relative over baseline+P1P3 (0.270/0.431).

4.3 ESS traces — block sizes 4–64

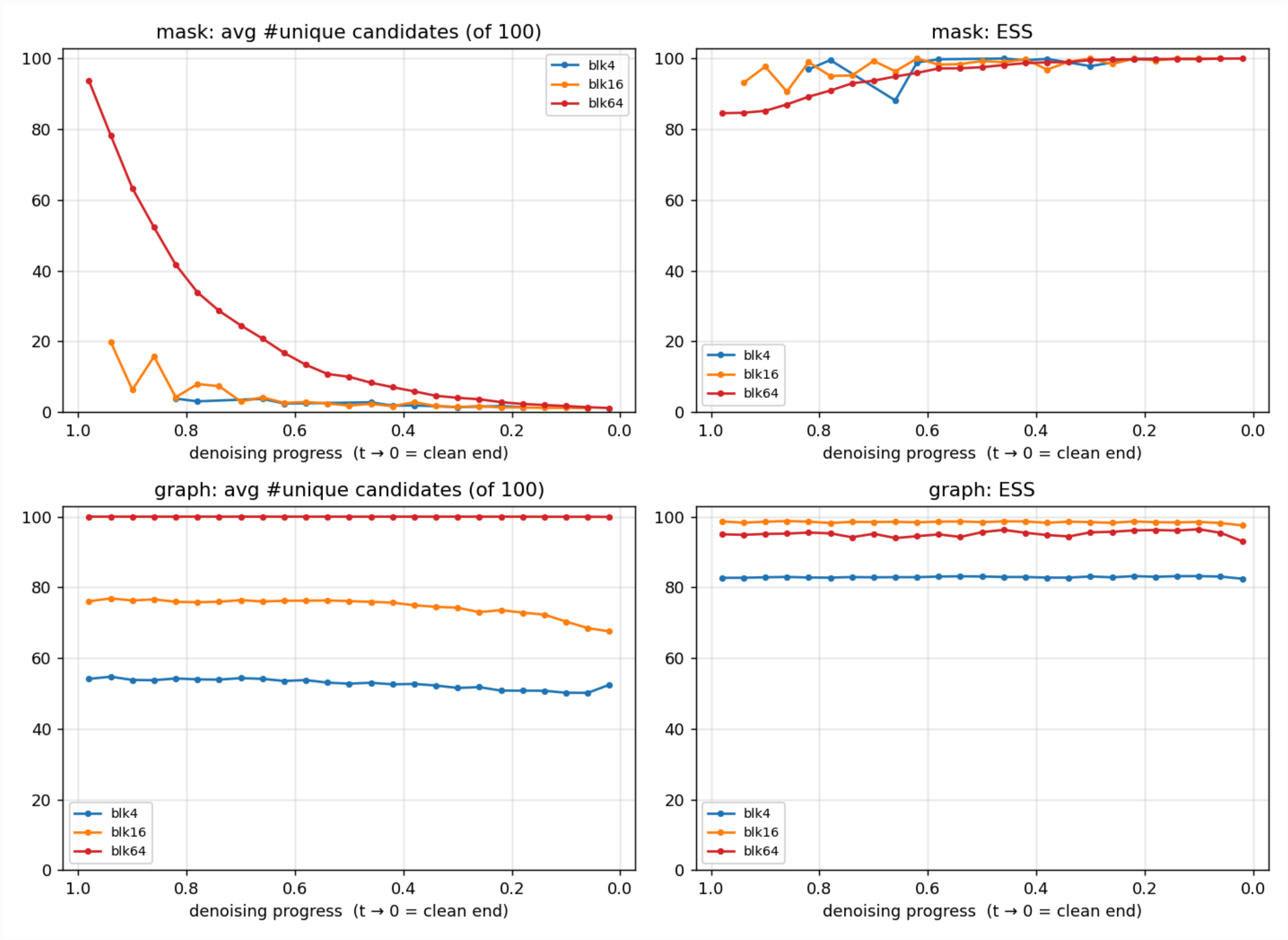
Reading ESS = $(\Sigma w)^2/\Sigma w^2 \in [1, N=100]$: a higher value (→100) means more candidates statistically survive, but in importance-weight guidance that means **weights are uniform = the disc cannot separate candidates = guidance is inert**. A lower value (e.g. 33) means weight concentrates on a few candidates = **the disc selects strongly = guidance is active**. The statistical "effective sample size" and the guidance "usefulness" run in opposite directions.



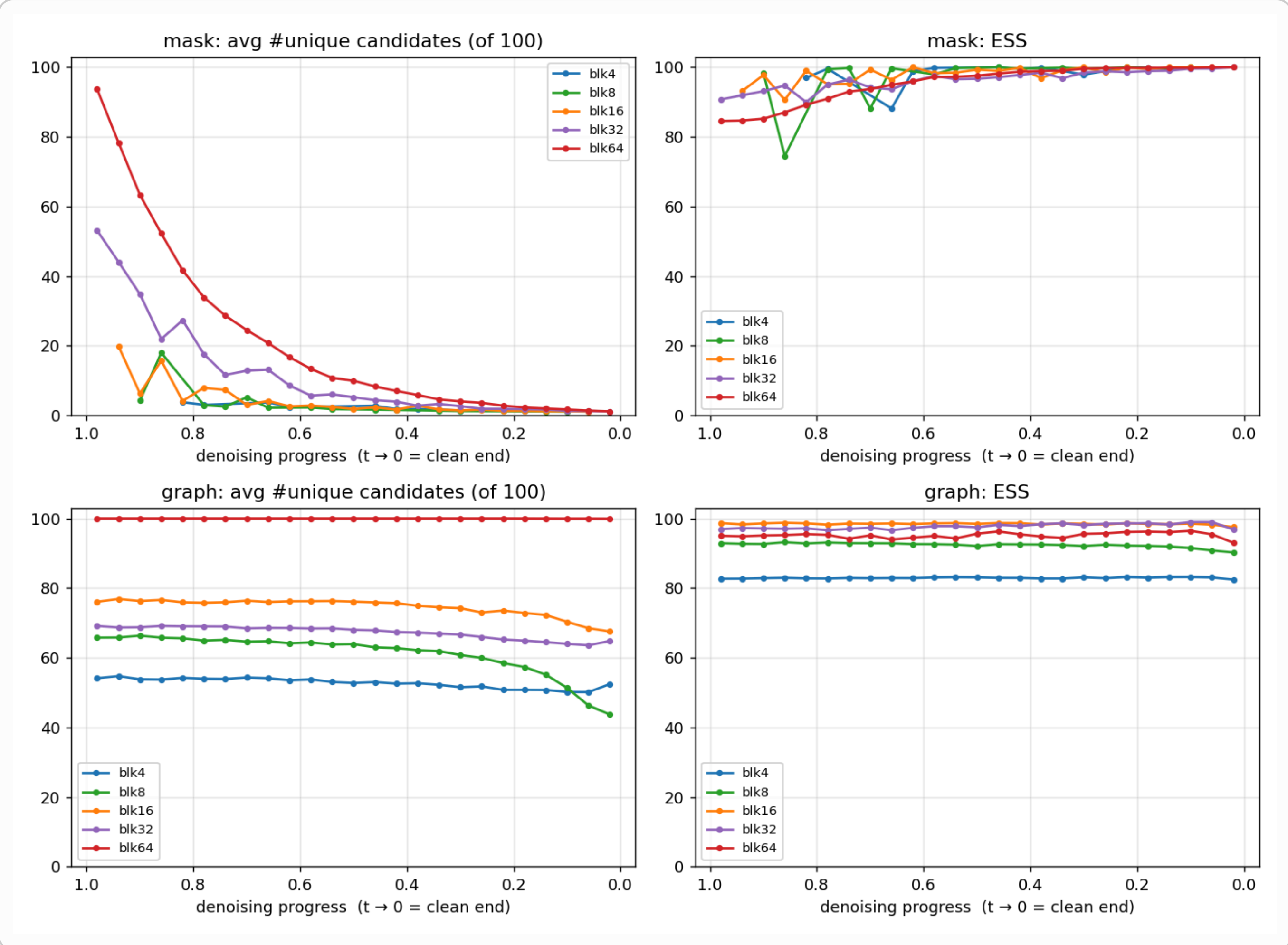
ESS per reveal/step (N=100, averaged over a 200-path batch, model-negative disc). **Left (mask)**: at every block size ESS is **lowest early in each block** (min 77–87) and converges to 100 as context fills — mask's irreversible reveals are most uncertain (diverse candidates) early, when guidance works; larger blocks linger near 100 longer. **Right (graph, full trace over generation)**: plotted like the mask panel — ESS is **low in the first blocks** (blk4 down to 33; first-5% mean 40) and **rises to ~100 as the path commits** in later blocks. Same "active-early" structure as mask, but the small-block graph kernel reaches far lower ESS (33 vs mask's 77–87), which is why guidance bites hardest there (§5.3); larger graph blocks start higher (94–97) with less headroom.

4.3b Candidate diversity per step — how many of the 100 candidates are unique

ESS measures weight *uniformity*, not the number of distinct candidate values. This panel measures the latter directly: average count of **unique** candidate blocks (of n_is=100) vs denoising progress (t→0 = clean end), on except_0 (120 pairs, seed 7, model-negative disc). **mask**: candidates collapse to ~1 as t→0 (larger blocks start diverse — blk64 ~94 — but all converge to a single sample by the clean end), which is exactly why mask ESS climbs to 100 (nothing left to reweight). **graph**: unique count stays high and flat (blk4 ~52, blk64 ~100) because the uniform limiting distribution keeps candidates diverse throughout — so graph guidance stays active. Note the two quantities diverge: graph blk64 has ~100 unique yet ESS ~94; mask blk64 at t≈1 has ~94 unique yet ESS ~84. A useful rule of thumb for the effective distinct contribution is min(ESS, #unique): #unique caps what reweighting can resolve, ESS caps how sharply it does so.



All block sizes (4 / 8 / 16 / 32 / 64), same diagnostic. The two intermediate sizes fill in a monotone trend: at the noisy start (t≈1) mask candidate diversity grows smoothly with block size (blk4 ~4 → blk8 ~4 → blk16 ~20 → blk32 ~53 → blk64 ~94), then every size collapses to ~1 unique by the clean end (t→0), so mask ESS rises to 100 across the board. graph unique count stays high and roughly flat throughout for every size (blk4 ~52 → blk64 ~100), with ESS ordered the same way — the uniform limiting distribution keeps candidates diverse to the end, so graph guidance stays active.



4.4 Reveal order × IW guidance — seen & unseen (model-negative disc, except_0)

The §2.4 left-to-right (l2r) reveal order applied **on top of IW guidance**, all block sizes, both discriminator regimes: **seen** (disc trained on except_0) and **unseen** (disc trained on except 1–99, evaluated zero-shot on except_0). Same reserved 1,000 pairs / seed 7 / model-negative disc as §4.1–§4.2. **Every cell is** `fh → l2r` (first-hitting → left-to-right); valid/EM are RAW, the last column is P1+P3; **bold** marks an l2r gain ≥2%p (arrival not bolded — its meaning is directional, see below). blk1/2 are order-invariant so only base is shown (blk1 exactly; blk2 has 2 within-block slots so a negligible difference).

Seen (disc = except_0):

blk	base valid	base arr	IW valid	IW arr	adj+IW valid	adj+IW arr	adj+IW P1P3 EM
1	0.306 → 0.312	0.994 → 0.995	—	—	—	—	—
2	0.264 → 0.263	0.956 → 0.967	—	—	—	—	—
4	0.266 → 0.282	0.966 → 0.977	0.331 → 0.333	0.965 → 0.970	0.390 → **0.708**	0.898 → 0.756	0.357 → **0.410**
8	0.231 → **0.281**	0.935 → 0.970	0.298 → **0.322**	0.950 → 0.962	0.346 → **0.752**	0.909 → 0.709	0.335 → **0.383**
16	0.222 → **0.266**	0.909 → 0.953	0.277 → **0.324**	0.915 → 0.949	0.327 → **0.743**	0.849 → 0.679	0.295 → **0.375**
32	0.160 → **0.253**	0.872 → 0.893	0.225 → **0.301**	0.832 → 0.879	0.255 → **0.748**	0.760 → 0.587	0.236 → **0.333**
64	0.141 → **0.235**	0.827 → 0.901	0.201 → **0.307**	0.854 → 0.879	0.229 → **0.828**	0.795 → 0.511	0.230 → **0.309**

Unseen (disc = except 1–99 → except_0):

blk	base valid	base arr	IW valid	IW arr	adj+IW valid	adj+IW arr	adj+IW P1P3 EM
1	0.306 → 0.312	0.994 → 0.995	—	—	—	—	—
2	0.264 → 0.263	0.956 → 0.967	—	—	—	—	—
4	0.266 → 0.282	0.966 → 0.977	0.329 → 0.331	0.963 → 0.971	0.390 → **0.705**	0.891 → 0.757	0.356 → **0.405**
8	0.231 → **0.281**	0.935 → 0.970	0.298 → **0.327**	0.951 → 0.962	0.342 → **0.750**	0.913 → 0.708	0.340 → **0.382**
16	0.222 → **0.266**	0.909 → 0.953	0.277 → **0.324**	0.917 → 0.946	0.326 → **0.742**	0.852 → 0.674	0.299 → **0.370**
32	0.160 → **0.253**	0.872 → 0.893	0.218 → **0.302**	0.832 → 0.884	0.249 → **0.748**	0.761 → 0.592	0.237 → **0.338**

64	0.141 → **0.235**	0.827 → 0.901	0.199 → **0.306**	0.857 → 0.879	0.224 → **0.828**	0.798 → 0.510	0.233 → **0.310**

- **adj+IW validity explodes under l2r** (seen blk64 0.229 → **0.828**, blk32 0.255 → 0.748; unseen nearly identical). The Lemma-3 adjacency-masked proposal only constrains a candidate when its **left neighbour is committed**; first-hitting reveals out of order so the mask is usually undefined, whereas l2r guarantees it at every reveal. Synergistic, not additive.
- **Arrival — the honest trade-off**: base and IW arrival stay high under both orders (0.83–0.98). Only **adj+IW** arrival *drops* under l2r (seen blk64 0.795 → 0.511): the strong validity constraint keeps every step legal but lets some paths overrun the destination. P3 endpoint-patch restores arrival to 1.00, so the net P1+P3 EM still rises (blk64 0.230 → 0.309).
- **IW and base also improve, growing with block size** (seen blk64 IW valid 0.201 → 0.307; base 0.141 → 0.235, matching §2.4).
- **Unseen mirrors seen** at every block and metric (generalization gap ≈ 0 holds under l2r too): e.g. blk64 adj+IW P1P3 EM seen 0.309 / unseen 0.310. l2r is a free, hyperparameter-free reveal-order change that compounds with IW + adj and transfers zero-shot.

5. Controlled comparison with the published baseline: D-CBG (Schiff et al.)

Plug-in port (their code logic verbatim; dcbg_plugin.py), **classifier architecture and training protocol fully matched** (same GraphSAGE + edge channels, same 1% data; model negatives use the same v2 uncond pools, noised their way). The only remaining differences are method-intrinsic: noisy-state input + per-position tilting (D-CBG) vs clean partial paths + joint candidate reweighting (IW). $\gamma = 4$ ($\gamma \le 2$ inert, v1 §7.1). Exact enumeration costs $|V| \approx$ 1.4k classifier calls per reveal (≈2.7 s/path); the cheap first-order variant (0.07 s/path) is **re-confirmed inert on v2** (blk4 valid 0.265 ≈ base 0.266); IW: 2.4k calls / 0.03 s/path.

Column definitions (explicit): **D-CBG (data/model)** = the **exact enumeration** at $\gamma = 4$ (NOT the first-order approximation; that variant is the separate row below) with the adjacency-aware classifier; negatives are normal real data / the v2 uncond pools respectively. **IW (model)** = importance-weight reweighting with the partial-path disc trained on model negatives (the same v2 pools). **adj+IW** = **exactly the same disc as IW (model)** plus Lemma 3 proposal masking only (i.e., yes — the model-data-trained disc). **Post-processing per column**: every *valid* column is RAW (pre-refinement); every *P1P3 EM* column is AFTER P1+P3. So each method shows raw validity beside its post-processed EM.

blk	D-CBG(data) valid	P1P3 EM	D-CBG(model) valid	P1P3 EM	IW(model) valid	P1P3 EM	adj+IW valid	P1P3 EM
4	0.301	0.295	0.309	0.302	0.331	0.309	0.390	0.357
8	0.257	0.275	0.272	0.275	0.298	0.294	0.346	0.335
16	0.249	0.263	0.258	0.252	0.277	0.268	0.327	0.295
32	0.185	0.231	0.185	0.203	0.225	0.216	0.255	0.236
64	0.163	0.222	0.173	0.209	0.201	0.211	0.229	0.230

D-CBG first-order (Taylor), measured at every block ($\gamma = 4$, model-negative classifier): blk4: 0.265 (base 0.266) · blk8: 0.233 (base 0.231) · blk16: 0.222 (base 0.222) · blk32: 0.161 (base 0.160) · blk64: 0.143 (base 0.141) — **inert at every block size** (all within noise of base). The gradient at the one-hot MASK embedding fails to approximate substitution effects. Wall-clock 0.074–0.090 s/path.

5.1 Wall-clock time (definition in §1.2)

method	blk4 s/path	blk64 s/path	note
base (no guidance)	0.009	0.007	—
IW (model)	0.026	0.066	~100 disc calls/reveal (3 micro-batches)
adj+IW	0.026	0.067	masking cost negligible
D-CBG exact (γ=4)	0.48–0.57	1.00–1.08	$|V| \approx$ 1,393 classifier calls/reveal
D-CBG first-order	0.074–0.27	—	1 fwd+bwd per reveal — but inert

Measured on one idle RTX 3090, batch 100. IW/base: a 200-pair timing loop (generation only, CUDA-synced); D-CBG exact: range between the 1,000-pair v3 runs and a 200-pair re-measurement; first-order: range between 1,000-pair (0.074) and 200-pair (0.27, startup-dominated). Reading: **IW is ~20× faster than exact D-CBG at blk4 and ~15× at blk64** — beyond the call-count gap (2.4k vs 33k), IW's calls are index-input forwards only, so they micro-batch efficiently. The first-order variant is fast but ineffective, hence dominated on the cost–quality frontier.

5.2 Cheating audit — where does the IW advantage come from

What is controlled: evaluation set (same except_0 reserved 1,000 pairs, same seeds), classifier architecture (same GraphSAGE + edge channels, bounded logit, label smoothing), training data (same 1%, splits, negative pools), backbones (same v2 checkpoints). D-CBG is reported at its best γ after a {1,2,4} sweep — tuning favours D-CBG. And exact D-CBG consumes MORE classifier information per reveal ($|V| \approx$ 1,393 evaluations vs IW's 100).

blk	base	placebo-IW (uniform w)	IW (real disc)	placebo-adj (masking only)	adj+IW (real)
4	0.266	0.267	**0.331**	0.331	**0.390**
16	0.222	0.211	**0.277**	0.281	**0.327**
64	0.141	0.143	**0.201**	0.179	**0.229**

- **Placebo control (ESS forced to 100.0)**: running the ENTIRE IW pipeline (100-candidate sampling → $\bar{x}_0$ averaging → reveal) with an untrained constant-logit disc lands **within noise of base** (blk4 0.267 vs 0.266). The IW gain is therefore not candidate averaging, sampling plumbing, or an implementation artifact — it is **entirely the discriminator signal**. Placebo-adj isolates the masking-only effect (+6–7%p), on top of which the real disc adds another +5–6%p — causally confirming §4's division of labour.
- **The one remaining asymmetry is method-intrinsic**: D-CBG's math requires noisy-state classification p(y|x_t,t); IW's (Lemma 1) is satisfied by a clean partial-path ratio. With matched architecture/data the classifiers train to comparable quality (val 0.73–0.75 / 0.82–0.83), but noisy-input classification is **structurally unlearnable on the uniform kernel** (§5.3). That is not cheating — it is the formulations differing, which is precisely the methodological contribution.
- Leakage checks: the diffusion never saw exceptional data; discs/classifiers train with the 1,000 eval pairs excluded; both methods share the same eval pairs. γ tuning and per-reveal classifier calls favour D-CBG.

5.3 The comparison on the GDP graph kernel (v2 graph blk4/64)

Block-structured (bfix) classifier retraining. The graph D-CBG classifier below was retrained with block-consistent corruption (committed blocks kept clean, only the current block noised; the full canvas fed to the classifier). This lifts classifier validation accuracy markedly — model-negative 0.84–0.91 across blk4–64; data-negative 0.68–0.71 for blk4–16 but decaying to ~chance (0.51/0.53) at blk32/64. Yet the first-order (Taylor) guided generation is essentially unchanged (≤0.03 on every metric): a well-trained classifier does not help because the first-order tilt is inert (§5 cheap-D-CBG finding) and the graph kernel's arrival is already near zero. The bottleneck is the approximation and the kernel, not classifier trainability — refuting the earlier "structurally untrainable" reading while leaving the IW > D-CBG ordering intact.

blk	method	valid raw	EM raw	arrival	ESS	P1P3 EM
4	base	0.393	0.138	0.137	—	—
	IW (model)	0.478	0.184	0.115	84.3	0.130
	adj+IW	**0.971**	**0.334**	0.042	81.5	**0.182**
	D-CBG fo (data)	0.377	0.141	0.133	—	0.102
	D-CBG fo (model)	0.397	0.118	0.124	—	0.083
8	base	0.427	0.069	0.064	—	—
	IW (model)	0.568	0.150	0.040	92.0	0.115
	adj+IW	**0.955**	**0.215**	0.015	90.1	**0.134**
	D-CBG fo (data)	0.402	0.074	0.064	—	0.050
	D-CBG fo (model)	0.406	0.075	0.066	—	0.051
16	base	0.498	0.114	0.030	—	—
	IW (model)	0.678	0.177	0.010	98.0	0.115
	adj+IW	**0.980**	**0.237**	0.002	97.7	**0.133**
	D-CBG fo (data)	0.504	0.103	0.024	—	0.060
	D-CBG fo (model)	0.482	0.116	0.028	—	0.073
32	base	0.462	0.089	0.029	—	—
	IW (model)	0.660	0.146	0.015	97.7	0.127
	adj+IW	**0.930**	**0.225**	0.004	96.3	**0.164**
	D-CBG fo (data)	0.474	0.092	0.024	—	0.068
	D-CBG fo (model)	0.461	0.079	0.024	—	0.067
64	base	0.473	0.108	0.034	—	—
	IW (model)	0.684	0.189	0.013	95.8	0.125
	adj+IW	**0.962**	**0.248**	0.008	95.8	**0.147**
	D-CBG fo (data)	0.465	0.122	0.042	—	0.085
	D-CBG fo (model)	0.469	0.121	0.036	—	0.082

- **Hypothesis confirmed: guidance bites much harder on the graph kernel with its uniform limiting distribution.** IW alone lifts valid by +8.5%p (blk4) / +21%p (blk64), and adj+IW essentially solves validity at **0.97/0.96** — versus +6.5%p on mask. ESS confirms the **kernel-level** reason (why graph > mask): on graph the weights genuinely differentiate — first-block min ESS ranges 33 (blk4) to ~94 (blk16) vs mask's ~77–87 floor that returns to 100 — so the high-entropy uniform proposal gives the disc candidates to select among. **But the within-graph block-size trend runs OPPOSITE to ESS**: the IW validity gain grows with block size (+8.5%p at blk4 → +21%p at blk64) while ESS is lowest at blk4. ESS is a per-step diagnostic (can the disc separate candidates now?) that does not integrate over reverse-step count or base-rate headroom, so it explains graph-vs-mask, not the block ordering. blk4's raw EM 0.334 / PC 0.512 is the best graph-kernel result on this benchmark. **Full-sweep confirmation (4–64)**: adj+IW valid is 0.93–0.98 at every graph block size (IW alone 0.48–0.68), essentially solving validity, while D-CBG first-order is inert under both negatives at every size (≈base) — the blk4/64 pattern holds throughout.
- **D-CBG has no viable variant on the graph kernel**: exact costs T×positions×|V| ≈ 8.9M calls (intractable); first-order is inert under both negatives (≈ base). The noisy-state classifier is at chance with data negatives (val 0.53–0.54; model negatives train to 0.87 but the approximation carries no signal).

- Cost: graph IW ≈ 0.4–0.9 s/path (T = 100 reverse × candidate scoring), D-CBG fo ≈ 0.5 s/path — comparable cost on graph, but only IW delivers an effect.
- **IW > D-CBG at every block size** (raw valid 10/10; blk4 0.331 vs 0.309), and the full stack adj+IW tops every cell. Our ingredients — the scenario-conditioned classifier and model negatives — port to D-CBG and improve it, yet the ordering persists.
- **Unseen transfer holds for both, with the IW lead intact**: blk4 D-CBG 0.306 / IW 0.325 (seen 0.309 / 0.331). Generalization is a property of the adjacency-conditioned classifier design (our contribution, portable).
- **Cost–quality frontier**: the cheap D-CBG (first-order) does not work and the working D-CBG (exact) is not cheap — IW dominates the region between (30–90× cheaper, no γ to tune, arrival preserved). On the uniform kernel their noisy-state classifier is structurally untrainable (chance level), making the gap decisive (v1 §7).

6. Conclusions and artifacts

- Small mask blocks (1–8) essentially solve the task; the residual defects (local skips, near-misses) are structurally repaired by P1/P3. The graph kernel is the opposite pole of the validity–arrival trade.
- The full exceptional-scenario system — **adj-masked proposals + model-negative partial-path discriminator (IW) + P1/P3** — is best on seen and unseen scenarios and beats the published baseline (D-CBG) in every matched configuration.
- Models: sets_model/*_v2_bd.pth (14; v1 under v1_archive/). Results: sets_res/, logs: sets_log/v3_*. Code: bd_models.py (plan_guided), bd_disc.py, dcbg_plugin.py, train_bd_disc.py, train_dcbg_classifier.py, eval_*.py, run_v3_all.sh.